

Dual Semantic Graph Construction

train-only priors · constrained adaptation

1 Data sources

statistics computed on training data

No validation/test statistics



raw + valid mask



wind-speed history

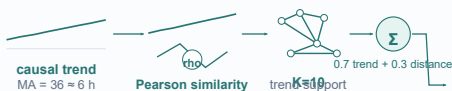


coordinates $r_i=(x_i, y_i)$

2 Semantic priors

Macro / m

trend semantics



Micro / mu

local-change semantics



Shared distance support

$$A_{d,ij} = \exp(-||r_i - r_j||^2 / \sigma^2)$$

distance

K=5

Macro prior

Micro prior

3 Prior-constrained adaptation

Independent node embeddings $E1, E2 \cdot \text{dim} = 10$

Reweight allowed edges only; unsupported links stay absent



same node set + same support

